



UNIMORE
UNIVERSITÀ DEGLI STUDI DI
MODENA E REGGIO EMILIA

Dipartimento di Ingegneria
“Enzo Ferrari”

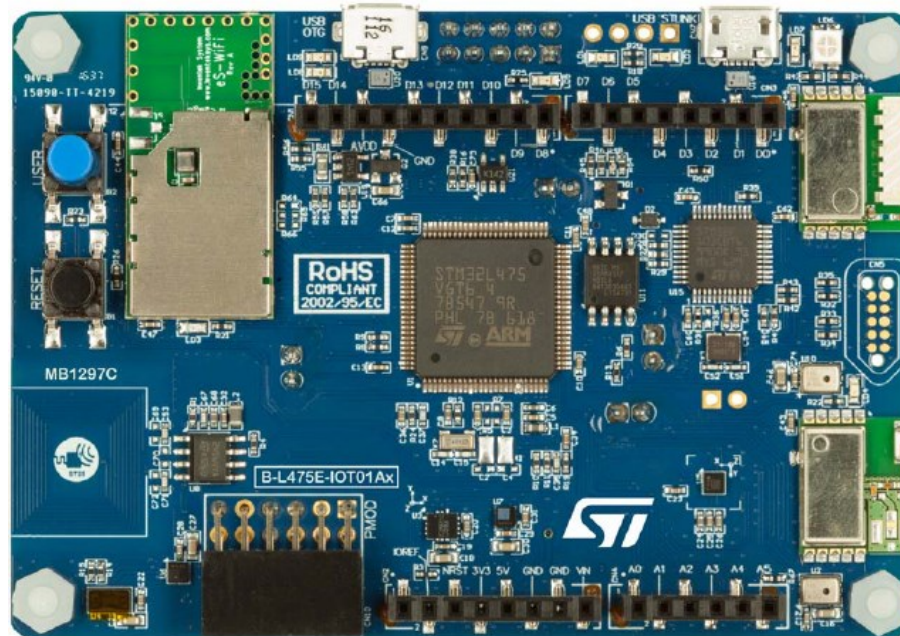
Fondamenti di elettronica

Esperienza di laboratorio 1: GPIO

Dott. Tommaso Zanotti

email: tommaso.zanotti@unimore.it

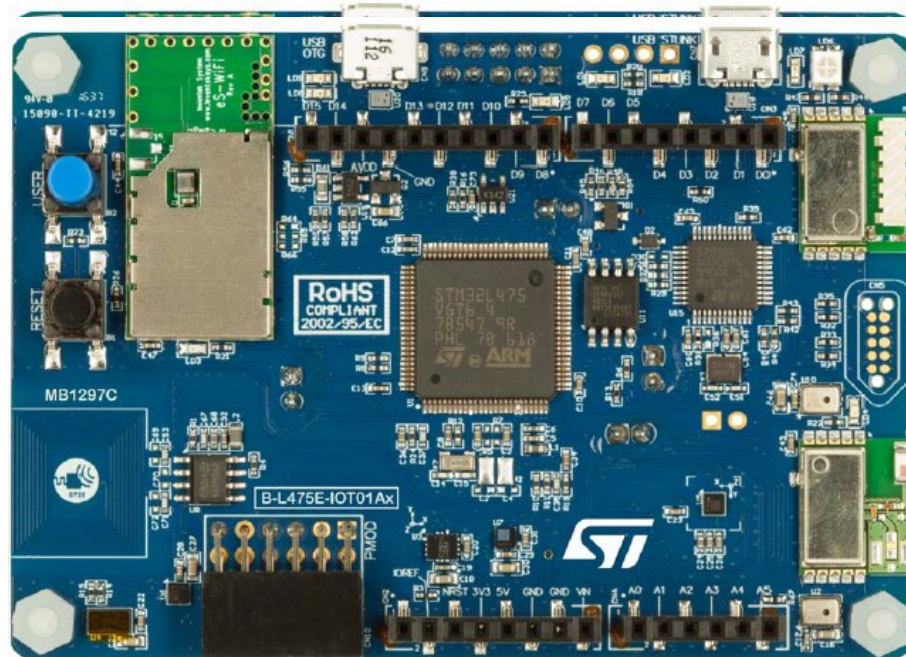
STM B-L475E-IOT01A



Documenti utili:

- [Descrizione compatta della board](#)
- [Descrizione più dettagliata](#)

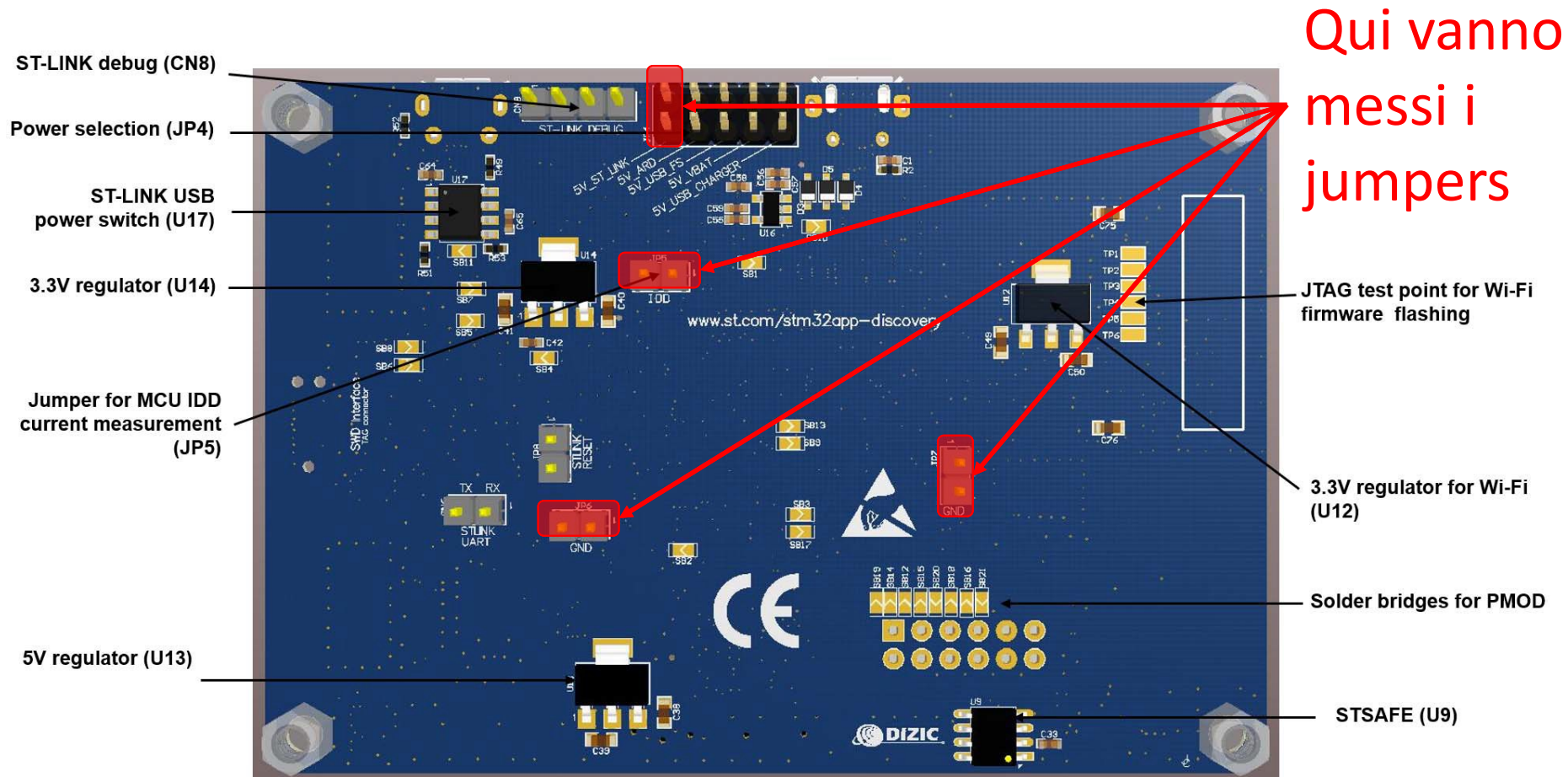
STM 32I475VG

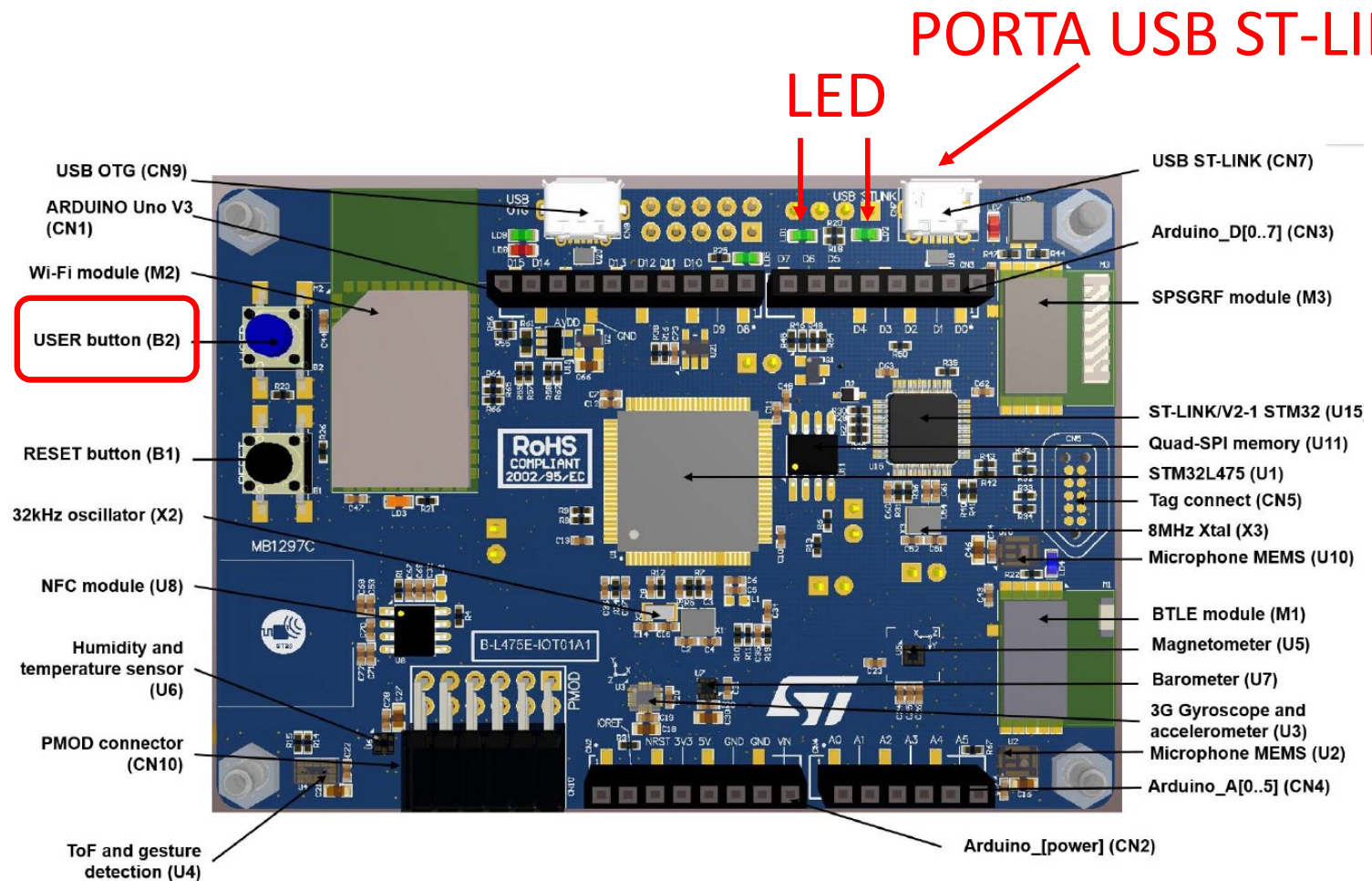


Documenti utili:

- [Descrizione compatta](#)
(Descrizione Funzionale, caratteristiche elettriche)
- [Reference Manual](#)
(Periferiche, configurazione dei registri)

Configurare correttamente i jumpers sulla scheda



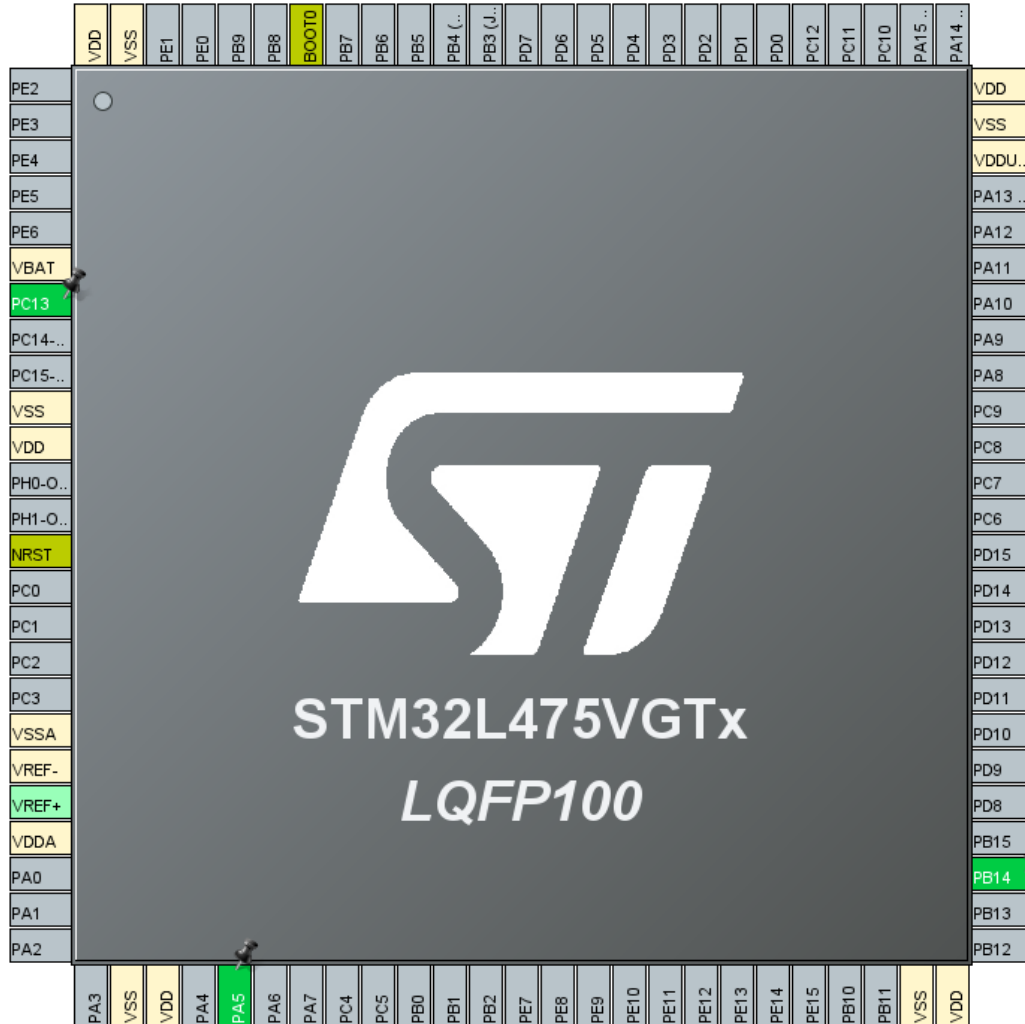


General Purpose Input Outputs

GPIO

- La funzionalità più semplice che può essere implementata da un microcontrollore è la capacità di **controllare il segnale digitale di pin** appartenenti al circuito integrato
- Un pin di **uscita digitale** può essere controllato per fornire in uscita un livello basso (0 V = valore logico "0"), o alto (V_{DD} = valore logico "1") di tensione
- Un **ingresso digitale** rileva se su un pin del microcontrollore sia presente un valore di tensione associato ad un valore logico "0" o "1"

GPIO Ports



Le **porte** sono raggruppamenti di **16** GPIO ciascuno.

I GPIO appartenenti a ciascuna porta sono controllati mediante gli stessi registri

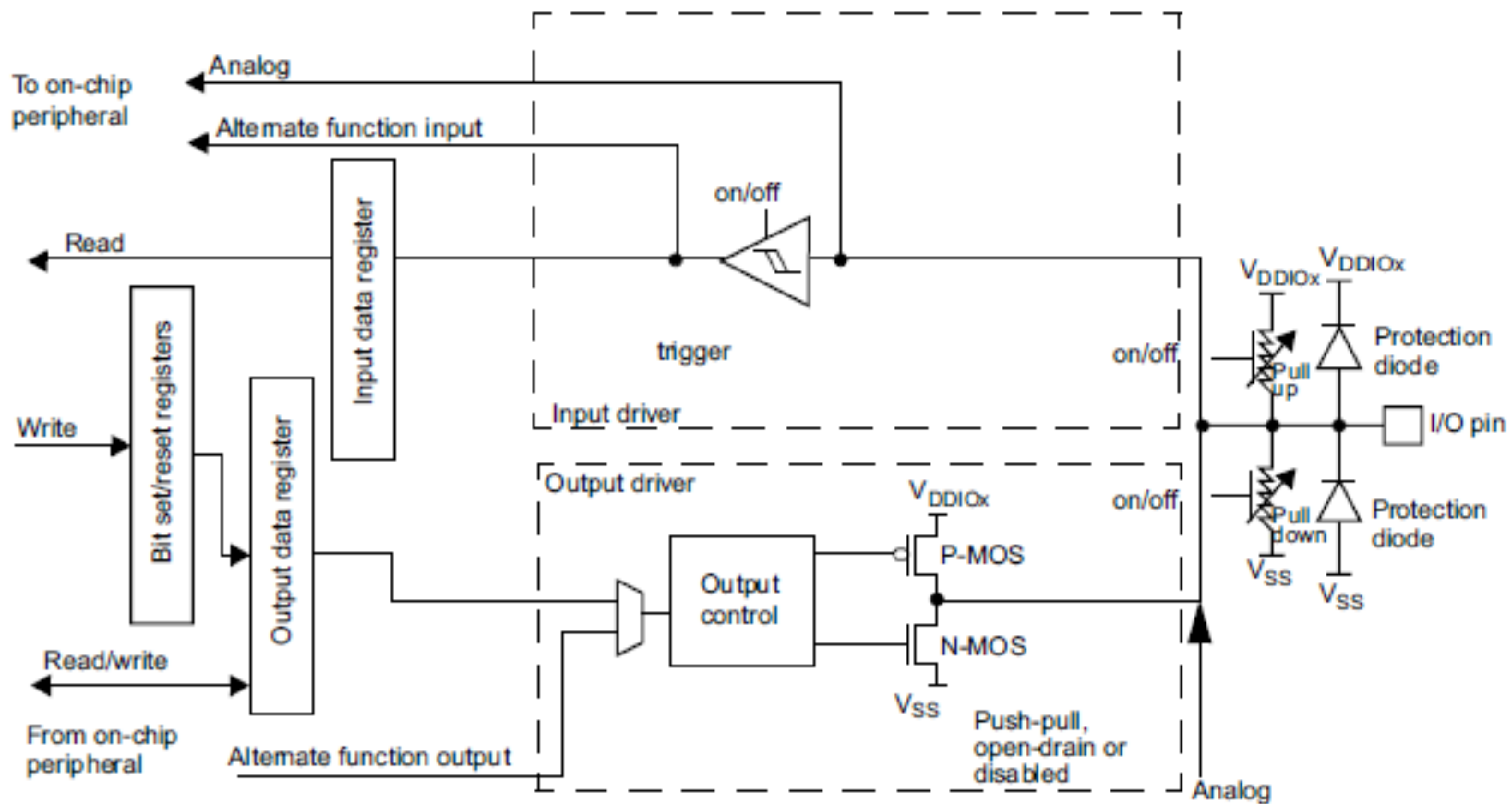
Il nostro microcontrollore mette a disposizione **5 porte** identificate dalle lettere **A, B, C, D, E**

Ogni singolo **bit** di ciascuna porta può essere **programmato** in modo **indipendente** dagli altri. Ad esempio la porta A può avere 6 pin configurati come ingressi e 10 pin come uscite

GPIO caratteristiche principali

- **Output states:** push-pull or open drain + pull-up/down
- Output data from **output data register** (GPIOx_ODR) or peripheral (alternate function output)
- **Speed selection** for each I/O
- **Input states:** floating, pull-up/down, analog
- Input data to **input data register** (GPIOx_IDR) or peripheral (alternate function input)
- **Locking mechanism** (GPIOx_LCKR) provided to freeze the I/O port configurations
- **Analog function**
- **Alternate function** selection registers
- **Fast toggle** capable of changing every **two clock cycles**
- **Highly flexible pin multiplexing** allows the use of I/O pins as GPIOs or as one of several peripheral functions

Struttura di porta di I/O



MS31476V1

Pag. 295 del documento "en.DM00347848_board_manual.pdf"

Input configuration

When the I/O port is programmed as input:

- The output buffer is disabled
- The Schmitt trigger input is activated
- The pull-up and pull-down resistors are activated depending on the value in the GPIOx_PUPDR register
- The data present on the I/O pin are sampled into the input data register every AHB clock cycle
- A read access to the input data register provides the I/O state

Output configuration

When the I/O port is programmed as output:

- The output buffer is enabled:
 - Open drain mode: A “0” in the Output register activates the N-MOS whereas a “1” in the Output register leaves the port in Hi-Z (the P-MOS is never activated)
 - Push-pull mode: A “0” in the Output register activates the N-MOS whereas a “1” in the Output register activates the P-MOS
- The Schmitt trigger input is activated
- The pull-up and pull-down resistors are activated depending on the value in the GPIOx_PUPDR register
- The data present on the I/O pin are sampled into the input data register every AHB clock cycle
- A read access to the input data register gets the I/O state
- A read access to the output data register gets the last written value

Alternate functions

The device I/O pins are **connected** to on-board peripherals/modules **through a multiplexer** that allows only one peripheral alternate function (AF) connected to an I/O pin at a time. In this way, there can be no conflict between peripherals available on the same I/O pin.

Each I/O pin has a multiplexer with up to **sixteen alternate function** inputs (AF0 to AF15) that can be configured through the GPIOx_AFRL (for pin 0 to 7) and GPIOx_AFRH (for pin 8 to 15) registers:

- After reset the multiplexer selection is alternate function 0 (AF0). The I/Os are configured in alternate function mode through GPIOx_MODER register.
- The specific alternate function assignments for each pin are detailed in the device datasheet.

In addition to this flexible I/O multiplexing architecture, each peripheral has alternate functions mapped onto different I/O pins to optimize the number of peripherals available in smaller packages.

Alternate function configuration

When the I/O port is programmed as alternate function:

- The output buffer can be configured in open-drain or push-pull mode
- The output buffer is driven by the signals coming from the peripheral (transmitter enable and data)
- The Schmitt trigger input is activated
- The weak pull-up and pull-down resistors are activated or not depending on the value in the GPIOx_PUPDR register
- The data present on the I/O pin are sampled into the input data register every AHB clock cycle
- A read access to the input data register gets the I/O state

Analog configuration

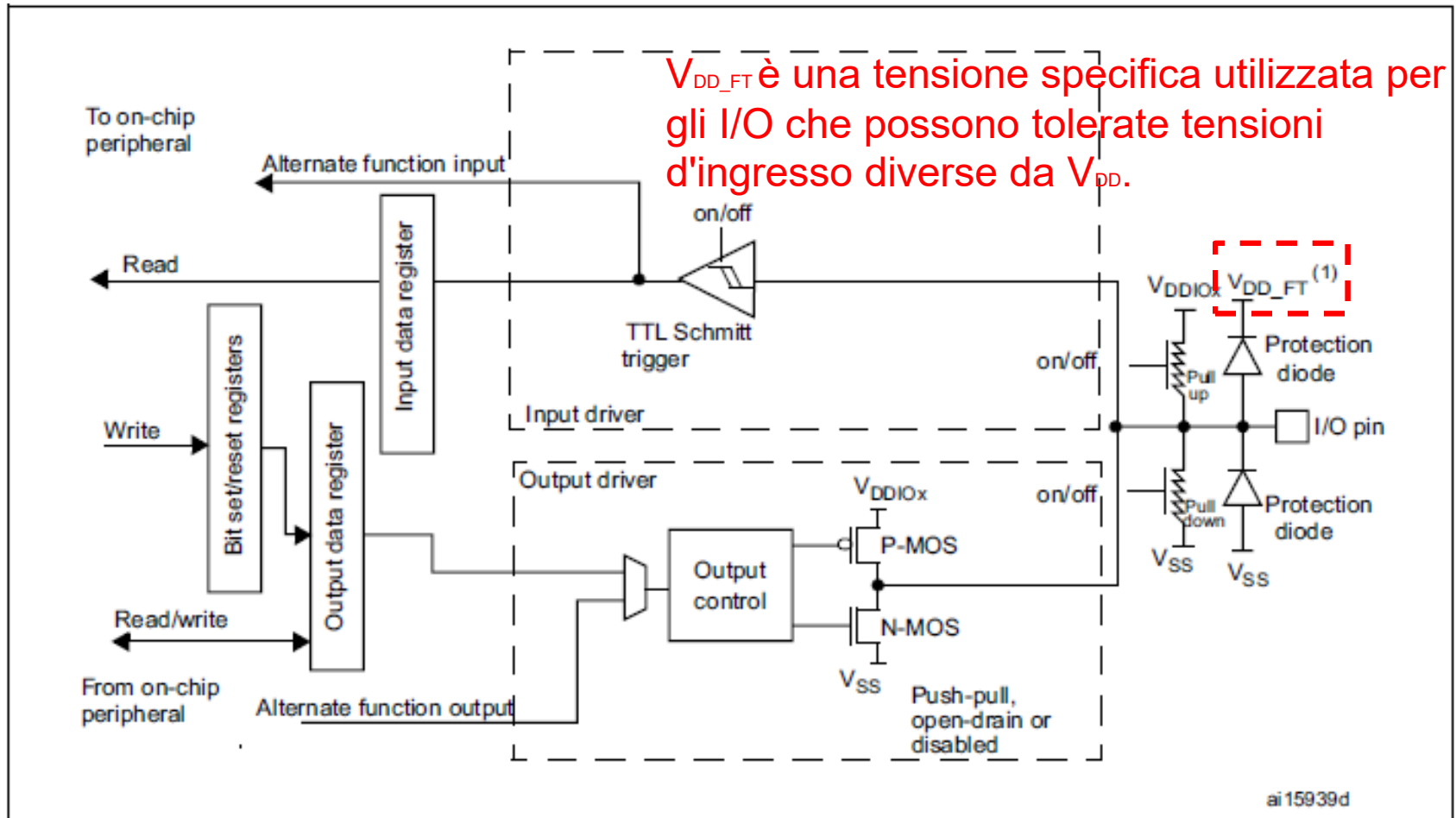
When the I/O port is programmed as analog configuration:

- The output buffer is disabled
- The Schmitt trigger input is deactivated, providing zero consumption for every analog value of the I/O pin. The output of the Schmitt trigger is forced to a constant value (0).
- The weak pull-up and pull-down resistors are disabled by hardware
- Read access to the input data register gets the value “0”

Additional functions

For the ADC, DAC, OPAMP, and COMP, **configure the desired I/O in analog mode** in the GPIOx_MODER register and configure the required function in the ADC, DAC, OPAMP, and COMP registers.

Struttura di porta di I/O 5V tolerant



Pag. 295 del documento "en.DM00347848_board_manual.pdf"

Struttura di porta di I/O 5V tolerant

Considerate sempre che tutti i pin del μC **NON siano 5V tolerant**, a meno che non abbiate effettivamente controllato sul datasheet.

La maggior parte dei pin **tollerano tensioni fino a 3.3V**, e se vengono collegati a segnali digitali da 5V possono **danneggiarsi irreparabilmente**.

Guardare documento "*stm32l475vg_Micro_Manual_Compatto.pdf*"

Corrente massima erogata/assorbita

Table 21. Current characteristics

Symbol	Ratings	Max	Unit
$\Sigma I_{V_{DD}}$	Total current into sum of all V_{DD} power lines (source) ⁽¹⁾	150	
$\Sigma I_{V_{SS}}$	Total current out of sum of all V_{SS} ground lines (sink) ⁽¹⁾	150	
$I_{V_{DD}(PIN)}$	Maximum current into each V_{DD} power pin (source) ⁽¹⁾	100	
$I_{V_{SS}(PIN)}$	Maximum current out of each V_{SS} ground pin (sink) ⁽¹⁾	100	
$I_{IO(PIN)}$	Output current sunk by any I/O and control pin except FT_f	20	mA
	Output current sunk by any FT_f pin	20	
	Output current sourced by any I/O and control pin	20	
$\Sigma I_{IO(PIN)}$	Total output current sunk by sum of all I/Os and control pins ⁽²⁾	100	
	Total output current sourced by sum of all I/Os and control pins ⁽²⁾	100	
$I_{INJ(PIN)}^{(3)}$	Injected current on FT_xxx, TT_xx, RST and B pins, except PA4, PA5	-5/+0 ⁽⁴⁾	
	Injected current on PA4, PA5	-5/0	
$\Sigma I_{INJ(PIN)} $	Total injected current (sum of all I/Os and control pins) ⁽⁵⁾	25	

Pag. 88 del documento "*stm32l475vg_Micro_Manual_Compatto.pdf*"

Registri di configurazione

Each general-purpose I/O port has:

- Four 32-bit configuration registers (GPIOx_MODER, GPIOx_OTYPER, GPIOx_OSPEEDR and GPIOx_PUPDR)
- Two 32-bit data registers (GPIOx_IDR and GPIOx_ODR)
- A 32-bit set/reset register (GPIOx_BSRR)
- A 32-bit locking register (GPIOx_LCKR)
- Two 32-bit alternate function selection registers (GPIOx_AFRH and GPIOx_AFRL)

Descrizione dei registri pag. 303 del documento
"en.DM00347848_board_manual.pdf"

Recap

- Not all GPIO Pin are 5v tolerant. Mostly they are 3.3v.
- You've to enable the clock for the GPIO you want to use regardless of the mode of operation
- When configuring a GPIO output pin, you can select the speed suitable for your application. Whether you need high-frequency switching on the I/O line or not
- The APB2 bus speed determines the sampling rate for all GPIO input configured pins
- You can disconnect any GPIO pin (make it high impedance) by setting it into input mode with Hi-Z
- You can freeze or lock the GPIO pins configuration after the system initialization if no change is intended using the locking mechanism
- Alternative function pins can be interchanged (remapped) internally so you can enjoy more freedom in routing these signals

Progetto 1: LED

Obiettivo: Far lampeggiare i LED presenti sulla scheda ad intervalli regolari, facendo in modo che il LED1 ed il LED2 siano l'uno il negato dell'altro.

To do:

1. Identificare i GPIO collegati ad i LED presenti sulla scheda
2. Creare il primo progetto con STM32CubeIDE
3. Provare le funzioni HAL per i GPIO

Progetto 2: Pulsante

Obiettivo: Far accendere il LED1 quando viene premuto il pulsante blu.

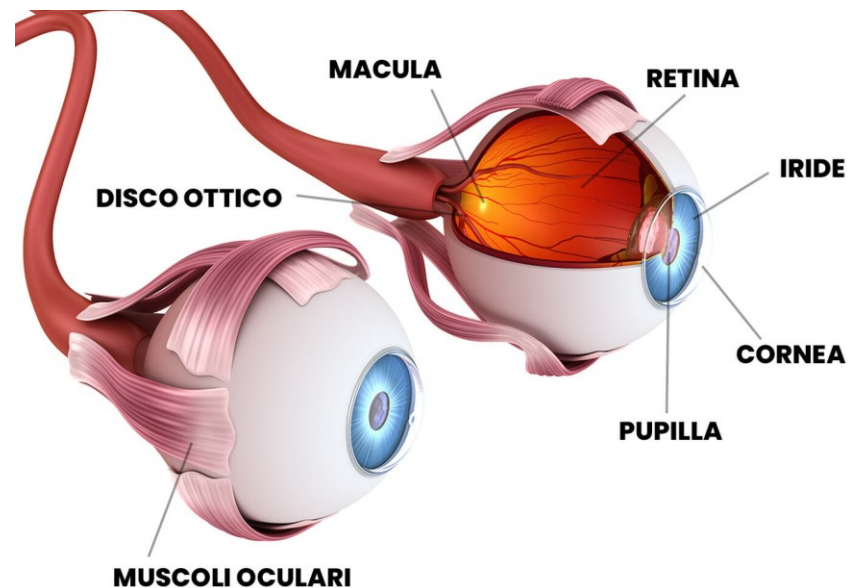
To do:

1. Identificare i GPIO collegato al pulsante
2. Provare le funzioni HAL per i GPIO

Fenomeno della persistenza visiva

E' un fenomeno per cui un'immagine viene trattenuta nel cervello per qualche frazione di secondo. Questo fenomeno è sfruttato in vari campi:

- Nel cinema
- Illuminazione
- ...



Progetto 3: «PWM» Fake

Obiettivo: Realizzare un controllo con tecnica PWM per il LED1. Valutare l'effetto di variazioni del periodo e del duty cycle.

To do:

1. Creare un nuovo progetto
2. Realizzare un controller PWM con la funzione HAL_Delay

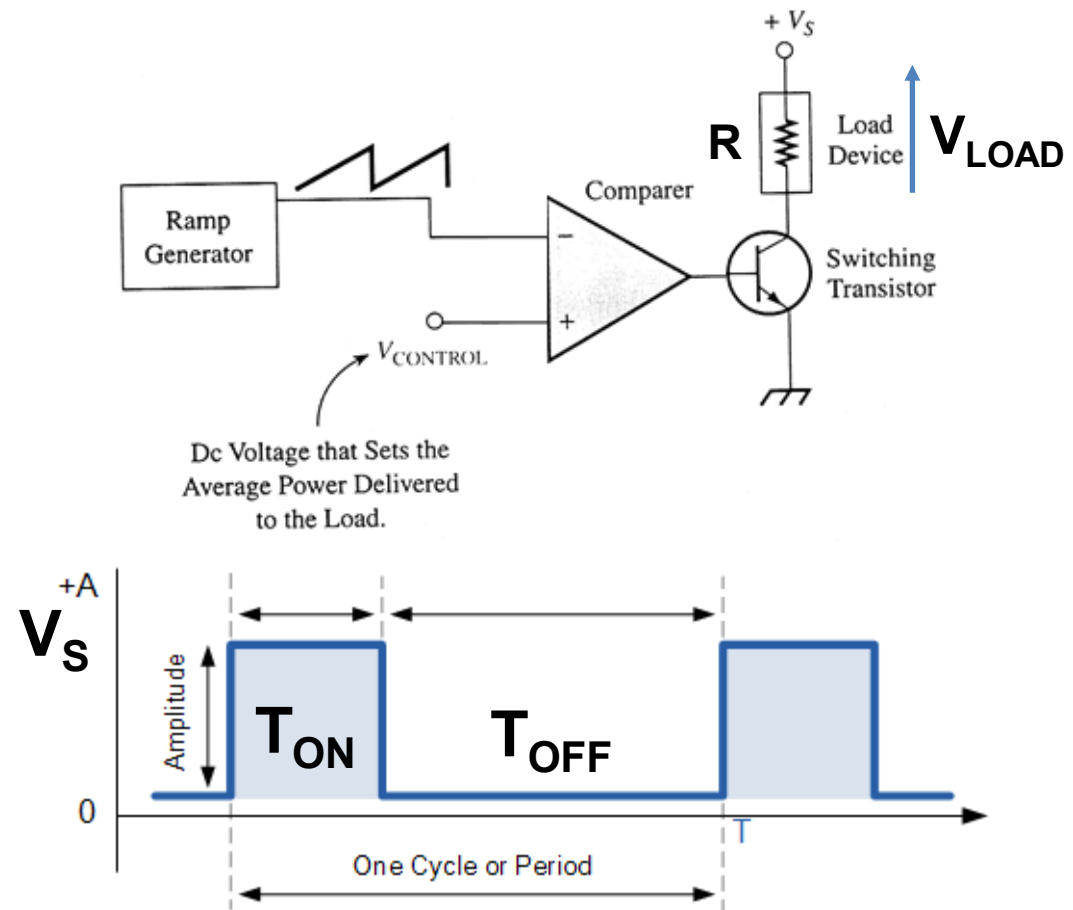
Pulse Width Modulation (PWM)

- Tecnica di modulazione che permette di variare la tensione (e potenza) media applicata su un carico

$$\overline{V_{LOAD}} = V_S \cdot D$$

$$\overline{P_{LOAD}} = D \cdot \frac{V_S^2}{R}$$

- Usata in moltissime applicazioni:
 - Controllo di motori DC
 - Dimmer
 - ...



$$Duty\ cycle\ (D) = \frac{T_{ON}}{T_{OFF} + T_{ON}}$$

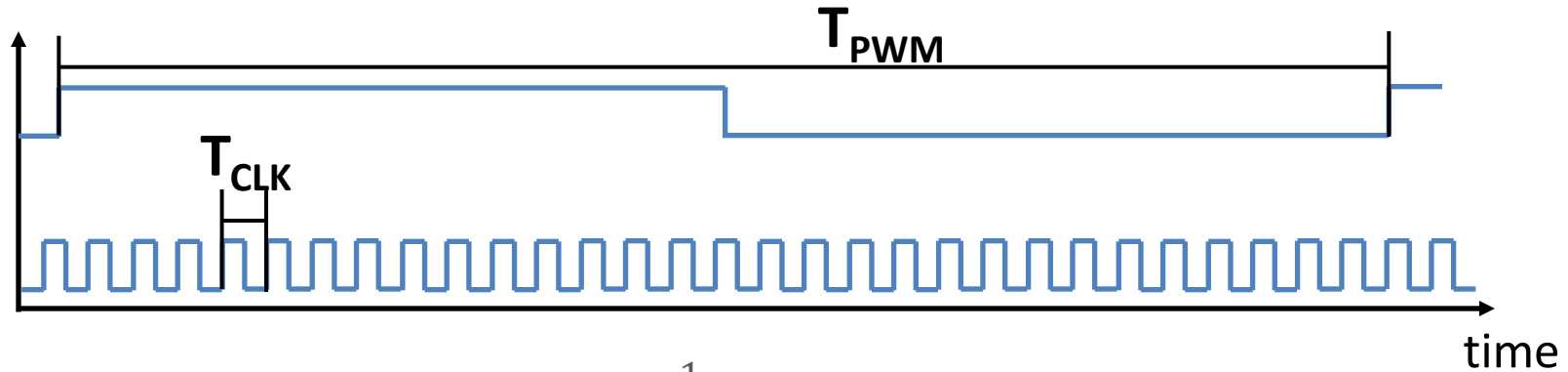
Progetto 4: PWM

Obiettivo: Realizzare un controllo con tecnica PWM per il LED1 sfruttando il general purpose timer TIM2 del microcontrollore. Configurare il TIM2 in modo che la frequenza del PWM ($1/T$) sia 25kHz.

To do:

1. Cercare nel manuale il TIMER 2 (TIM2)
2. Creare un nuovo progetto
3. Configurare correttamente il timer
4. Variare periodicamente l'intensità luminosa del LED1

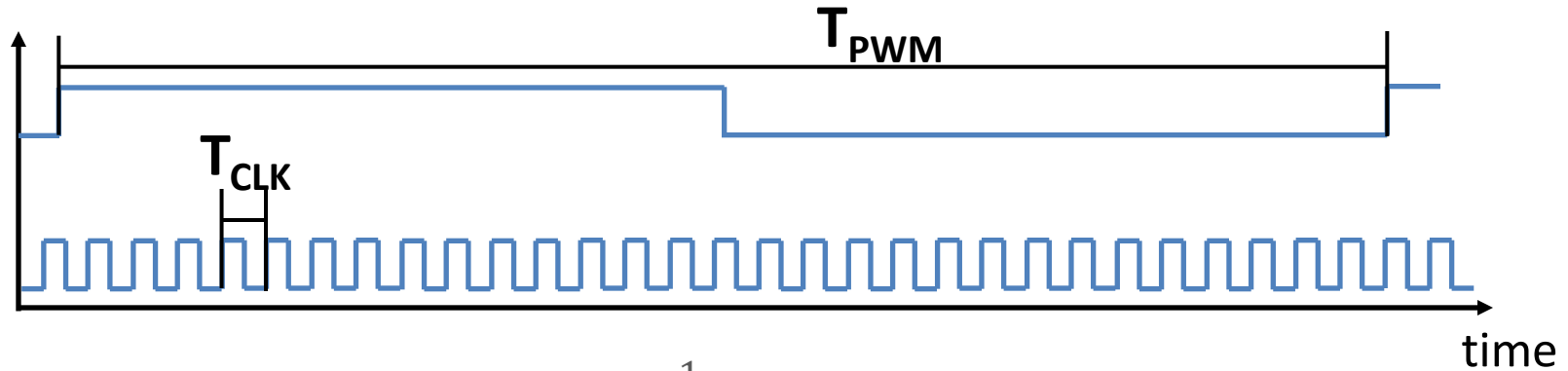
Come calcolare il numero massimo del contatore



- $f_{CLK} = 80MHz \Rightarrow T_{CLK} = \frac{1}{8 \cdot 10^7}$
- $f_{PWM} = 25kHz \Rightarrow T_{PWM} = \frac{1}{25 \cdot 10^3}$

Un periodo del PWM quanti periodi di CLK contiene?

Come calcolare il numero massimo del contatore



- $f_{CLK} = 80MHz \Rightarrow T_{CLK} = \frac{1}{8 \cdot 10^7}$
- $f_{PWM} = 25kHz \Rightarrow T_{PWM} = \frac{1}{25 \cdot 10^3}$
- $\Rightarrow N_{MAX} = \frac{T_{PWM}}{T_{CLK}} = \frac{8 \cdot 10^7}{25 \cdot 10^3} = 3200$